

FAIRCHILD AFB, WA, USA

WMO: 727855

Lat: 47.633N

Lon: 117.650W

Elev: 750

StdP: 92.63

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24114

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-14.2	-11.3	-18.5	0.8	-12.5	-15.5	1.1	-9.0	12.6	4.2	11.3	1.8	3.2	30	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	33.7	16.7	32.0	16.2	30.0	15.8	18.2	29.1	17.3	28.6	16.5	27.5	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	11.7	19.6	13.6	10.7	18.6	12.4	9.8	18.3	54.0	29.0	51.2	28.5	48.7	27.6	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.2	8.2	-18.3	36.7	4.6	1.7	-21.6	37.9	-24.2	38.9	-26.8	39.9	-30.1	41.2		
(5)			-17.8	19.7	4.9	1.2	-21.3	20.6	-24.2	21.3	-27.0	22.0	-30.6	22.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-1.8	-0.1	4.1	7.9	12.5	16.1	21.3	20.6	15.6	8.2	2.4	-1.9	(6)
(7)		DBStd	9.00	4.56	4.35	3.67	3.53	4.01	3.91	3.60	3.60	4.07	3.92	4.14	4.39	(7)
(8)		HDD10.0	1624	367	282	184	84	21	2	0	0	5	80	229	370	(8)
(9)		HDD18.3	3726	626	516	442	314	185	90	16	19	99	313	478	628	(9)
(10)		CDD10.0	1180	0	0	1	20	98	184	349	328	174	26	1	0	(10)
(11)		CDD18.3	240	0	0	0	0	4	22	106	89	18	1	0	0	(11)
(12)		CDH23.3	3298	0	0	0	3	78	324	1433	1163	288	7	0	0	(12)
(13)		CDH26.7	1338	0	0	0	0	12	102	641	496	86	0	0	0	(13)
(14)	Wind	WSAvg	3.8	3.7	4.0	4.4	4.3	4.0	3.8	3.4	3.4	3.7	3.9	3.8	(14)	
(15)	Precipitation	PrecAvg	397	47	35	37	30	38	29	15	16	17	26	53	56	(15)
(16)		PrecMax	641	114	83	97	65	93	72	59	48	48	83	122	125	(16)
(17)		PrecMin	267	10	1	6	2	5	3	0	0	0	1	5	14	(17)
(18)		PrecStd	74	23	19	18	17	21	16	16	14	13	20	29	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	11.5	17.2	23.6	28.2	33.0	36.4	35.6	32.4	24.0	14.0	10.1	(19)
(20)			MCWB	7.3	7.1	9.1	12.9	15.1	17.0	17.2	16.8	15.8	13.6	10.1	8.4	(20)
(21)		2%	DB	6.4	8.9	14.2	20.2	25.6	29.5	34.0	33.3	29.0	20.0	11.5	7.2	(21)
(22)			MCWB	5.0	6.4	8.2	11.3	14.3	16.0	17.0	16.4	14.8	11.5	9.1	6.0	(22)
(23)		5%	DB	4.9	7.1	12.3	17.4	23.1	27.1	32.4	31.7	26.8	17.3	9.8	5.1	(23)
(24)			MCWB	3.9	4.9	7.3	9.7	13.2	15.1	16.3	16.1	14.2	10.5	7.7	4.2	(24)
(25)		10%	DB	3.5	5.4	10.1	14.9	20.8	24.7	30.4	29.7	24.1	14.8	8.0	3.6	(25)
(26)			MCWB	2.8	3.8	6.2	8.5	12.1	14.1	16.0	15.8	13.3	9.2	6.3	2.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	8.4	10.9	14.0	16.4	18.7	20.2	18.9	17.2	14.8	11.6	8.4	(27)
(28)			MCDWB	7.9	9.9	13.9	21.3	23.1	29.5	29.3	29.9	28.5	20.6	12.6	9.5	(28)
(29)		2%	WB	5.4	6.9	9.2	12.0	15.2	17.0	18.6	17.8	15.7	12.9	9.7	6.1	(29)
(30)			MCDWB	6.2	8.4	12.8	17.8	23.3	26.8	29.1	29.4	26.2	17.8	11.2	7.0	(30)
(31)		5%	WB	4.0	5.3	7.9	10.5	14.1	15.9	17.7	16.9	14.8	11.4	8.0	4.3	(31)
(32)			MCDWB	4.7	6.6	11.0	16.1	21.4	24.9	29.0	28.6	24.7	15.8	9.1	4.9	(32)
(33)		10%	WB	2.8	4.0	6.9	9.2	12.9	14.9	16.8	16.2	13.9	9.9	6.8	2.9	(33)
(34)			MCDWB	3.4	5.2	9.7	14.1	19.3	23.1	28.1	27.6	22.9	14.0	7.8	3.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.2	7.0	9.5	11.0	12.1	12.7	15.2	14.9	13.8	10.8	7.0	5.1	(35)
(36)			MCDBR	6.1	8.7	12.5	14.6	15.4	15.9	17.4	17.4	17.2	14.2	8.8	6.1	(36)
(37)			MCWBR	5.0	6.1	7.5	7.4	6.6	6.0	5.9	6.3	7.0	7.3	6.2	5.3	(37)
(38)		5% WB	MCDWB	5.9	7.7	10.8	13.0	13.8	14.5	15.5	15.7	15.3	12.3	7.7	5.8	(38)
(39)			MCWBR	5.1	5.8	7.1	7.2	6.3	5.9	5.7	6.1	6.7	6.9	6.0	5.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.270	0.278	0.290	0.310	0.324	0.326	0.325	0.333	0.311	0.298	0.278	0.266	(40)	
(41)		taud	2.503	2.544	2.570	2.492	2.474	2.496	2.500	2.458	2.512	2.547	2.559	2.507	(41)	
(42)		Ebn at Noon	834	906	944	946	939	935	932	910	904	866	819	794	(42)	
(43)		Edh at Noon	61	73	83	99	105	104	102	102	87	72	58	54	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.04	2.00	3.08	4.61	5.88	6.34	7.39	6.14	4.39	2.62	1.25	0.83	(44)	
(45)		RadStd	0.13	0.27	0.34	0.42	0.46	0.53	0.39	0.40	0.39	0.29	0.14	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	-0.40	N/A	N/A	N/A	+92	+45

Nomenclature: See separate page